Antihypertensive Pharmacology: Lecture 1 Examination

Part I: Multiple Choice Questions (MCQs)

Section 1: Basics and Classification of Hypertension

1. The physiological formula for Blood Pressure (BP) is:
   - A. BP = SV x HR
   - B. BP = CO x PR
   - C. BP = EDV x PR
   - D. BP = CO / PR
2. According to JNC 7, "Stage 1 Hypertension" is defined as a Systolic BP of:
   - A. 120 – 139 mm Hg
   - B. 140 – 159 mm Hg
   - C. 160 mm Hg or higher
   - D. Less than 120 mm Hg
3. Which of the following is considered a "receiving" phase where the heart muscle is resting and refilling with blood?
   - A. Systolic pressure
   - B. Diastolic pressure
   - C. Preload pressure
   - D. Afterload resistance

Section 2: Etiology and Complications

4. Approximately what percentage of hypertension cases are classified as "Primary (Essential)" or idiopathic?
   - A. 10-20%
   - B. 50%
   - C. 80-90%
   - D. 100%

5. Which of the following is a secondary cause of hypertension related to an endocrine disease?
   - A. Pheochromocytoma
   - B. High salt intake
   - C. Smoking
   - D. Middle age
6. Chronic hypertension is often called a "silent killer" because it can lead to:
   - A. Myocardial Infarction
   - B. Brain strokes (ischemic or hemorrhagic)
   - C. Retinopathy
   - D. All of the above

Section 3: Physiological Control Mechanisms

7. The amount of blood pumped by the left ventricle in one single beat is called:
   - A. Cardiac Output
   - B. Stroke Volume (SV)
   - C. End Diastolic Volume
   - D. Peripheral Resistance
8. Which of the following substances acts as a natural vasoconstrictor produced by the vascular endothelium?
   - A. Nitric oxide
   - B. Prostacycline (PGI2)
   - C. Endothelin
   - D. Bradykinin
9. Which drug acts specifically as an endothelin receptor antagonist?
   - A. Clonidine
   - B. Bosentan
   - C. Prazosin

- D. Atenolol

Section 4: Sympatholytic Drugs (Centrally Acting & Alpha Blockers)

10. The mechanism of action for Clonidine is:
    - A. $α1$ receptor blockade
    - B. Presynaptic $α2$ agonism
    - C. $β1$ receptor blockade
    - D. Ganglion blockade
11. Which drug is preferred for treating hypertension during pregnancy?
    - A. Clonidine
    - B. $α$-Methyl dopa
    - C. Propranolol
    - D. Guanethidine
12. A common side effect of Prazosin, often requiring combination with Propranolol, is:
    - A. Hemolysis
    - B. Dry nasal mucosa
    - C. Orthostatic hypotension with reflex tachycardia
    - D. Rebound hypertension

Section 5: Beta-Blockers and Combined Blockers

13. Which beta-blocker is classified as "Cardioselective" but also produces a vasodilating effect via Nitric Oxide (NO)?
    - A. Propranolol
    - B. Atenolol
    - C. Nebivolol
    - D. Timolol
14. Beta-blockers are generally considered more effective in:
    - A. Black patients and elderly patients

  - B. White patients and young patients
  - C. Diabetic patients with asthma
  - D. Acute heart failure patients

15. Abrupt withdrawal of beta-blockers in patients with ischemic heart disease can induce:
    - A. Hallucinations
    - B. Impotence
    - C. Angina or Myocardial Infarction
    - D. Hemolysis
16. Which drug is a combined $α$ and $β$ blocker used in hypertensive crises and pheochromocytoma?
    - A. Labetalol
    - B. Bisoprolol
    - C. Terazosin
    - D. Clonidine

Part II: Essay & Short Answer Questions

1. Define Hypertension and provide the classification table for Blood Pressure categories based on JNC 7.
2. List four potential complications of untreated chronic hypertension.
3. Compare Primary (Essential) and Secondary Hypertension in terms of etiology and percentage of occurrence.
4. Explain the mechanism of the Baroreflex control of blood pressure when BP increases.
5. Briefly describe the mechanism of action and the specific clinical use for $α$-Methyl dopa.
6. List five therapeutic uses for $β$-blockers other than hypertension.
7. Identify the contraindications for using $β$-blockers.

Answer Key

Part I: MCQs

1. B | 2. B | 3. B | 4. C | 5. A | 6. D | 7. B | 8. C | 9. B | 10. B | 11. B | 12. C | 13. C | 14. B | 15. C | 16. A

Part II: Essay Hints

1. Definition: Sustained increase in BP (>130/80). Table: Normal (<120/80), Pre-HTN (120-139/80-89), Stage 1 (140-159/90-99), Stage 2 (≥160/100).
2. Complications: Brain stroke, Heart failure, Myocardial Infarction, Renal failure, and Retinopathy.
3. Primary: 80-90%, Idiopathic (salt, stress, smoking). Secondary: 10-20%, caused by endocrine disease, renal disease, or drugs.
4. Baroreflex: Increase in BP → distension of aorta/carotid → stimulates baroreceptors → impulses to vasomotor center → decreased Sympathetic and increased Parasympathetic stimulation.
5. α-Methyl dopa: Converted to α-methyl NE, displaces NE from vesicles (less $\alpha1$ effect) and stimulates $\alpha2$ receptors (decreased release). Used in pregnancy and renal-complicated hypertension.
6. Uses: Angina/MI, Supraventricular arrhythmias, Migraine prophylaxis, Glaucoma, Hyperthyroidism, and Anxiety.
7. Contraindications: Congestive heart failure (-ve inotropic), Bronchial asthma (non-selective), and Diabetes (especially Propranolol).